Federated Learning Project Report

Centralized Baseline vs. FedAvg on FashionMNIST and CIFAR-10

Based on: McMahan et al., "Communication-Efficient Learning of Deep Networks from Decentralized Data," AISTATS 2017 (arXiv:1602.05629)

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1. Introduction

Federated Learning (FL) is a distributed machine learning paradigm that enables model training across multiple decentralized clients, each holding a private local dataset, without ever sharing raw data with a central server. Instead of uploading data, each client computes model updates locally and only transmits these updates to the server for aggregation. This approach directly addresses privacy concerns in real-world deployments such as mobile devices, hospitals, and financial institutions.

This project implements and evaluates the **Federated Averaging (FedAvg)** algorithm proposed by McMahan et al. (2017) on two image classification benchmarks: **FashionMNIST** and **CIFAR-10**. Five simulated clients are used, with the main process acting as the server. The key objectives are:

- Establish a centralized training baseline (upper bound) for each dataset.
- Implement FedAvg with 5 clients under non-IID, non-overlapping data partitions.
- Quantify the accuracy gap and variance between centralized and federated training.
- Analyze the effect of dataset complexity and data heterogeneity on FedAvg.

The FedAvg Algorithm: In each communication round, the server broadcasts the current global model weights to a fraction C of clients. Each selected client runs E local epochs of SGD on its private data, then sends updated weights back. The server aggregates these updates via a weighted average (proportional to each client's dataset size) to produce the new global model. This repeats for T communication rounds.

2. Experimental Setup

2.1 Datasets

Dataset	Train Size	Test Size	Input Shape	Classes	Normalization
FashionMNIST	60,000	10,000	28×28 (grayscale)	10	mean=0.5, std=0.5
CIFAR-10	50,000	10,000	32×32×3 (RGB)	10	Per-channel mean/std

Batch size: 64 for all experiments. The test set is held centrally at the server.

2.2 Model Architectures

Dataset	Architecture	Loss	Optimizer	LR
FashionMNISTML	Flatten → Linear(784,128)+ReLU → Linear(128,10)	CrossEntropy	Adam	0.001
CIFAR-10	Conv(3,32,3)+ReLU+MaxPool → Conv(32,64,3)+ReLU+MaxPool → FC(4096,128)+ReLU → FC(128,10)	CrossEntropy	Adam	0.001

2.3 Federated Learning Configuration

Parameter	Value / Description
Number of clients (K)	5
Client fraction (C)	0.95 — effectively all clients participate each round
Local epochs (E)	1 epoch per communication round
Communication rounds	20 rounds per run
Number of runs	5 independent runs (fresh model each time)
Aggregation rule	Weighted average by local dataset size (FedAvg)
Non-IID split — FashionMNIST	Primary fraction = 0.85 (moderate skew)
Non-IID split — CIFAR-10	Primary fraction = 0.95 (strong skew)

2.4 Non-IID Data Partitioning Strategy

Training data is split across clients using a **by-label skew** strategy, ensuring no overlap between client datasets. Each of the 10 classes is assigned a "primary" client as follows: classes 0,1 → Client 0; classes 2,3 → Client 1; classes 4,5 → Client 2; classes 6,7 → Client 3; classes 8,9 → Client 4.

The primary client receives *primary_fraction* of all samples for its two assigned classes. The remaining samples are distributed round-robin to the other four clients. This creates a realistic scenario where each client is an expert on its dominant classes but has limited exposure to others — a hallmark of real-world federated settings (e.g., different hospitals specializing in different conditions).

3. Question 1 — Centralized Baseline: FashionMNIST

3.1 Setup

The two-layer MLP is trained on the complete FashionMNIST training set (60,000 samples) for 20 epochs. The experiment is repeated 5 independent times with fresh random initializations. Both training accuracy and test accuracy are recorded at every epoch. This represents the performance upper bound: all data is available, no communication overhead, and no model averaging artifacts.

3.2 Results

Metric	Value
Average Final Test Accuracy (5 runs)	88.22%
Variance of Final Test Accuracy	0.0830

3.3 Training and Test Accuracy Curves (All 5 Runs)

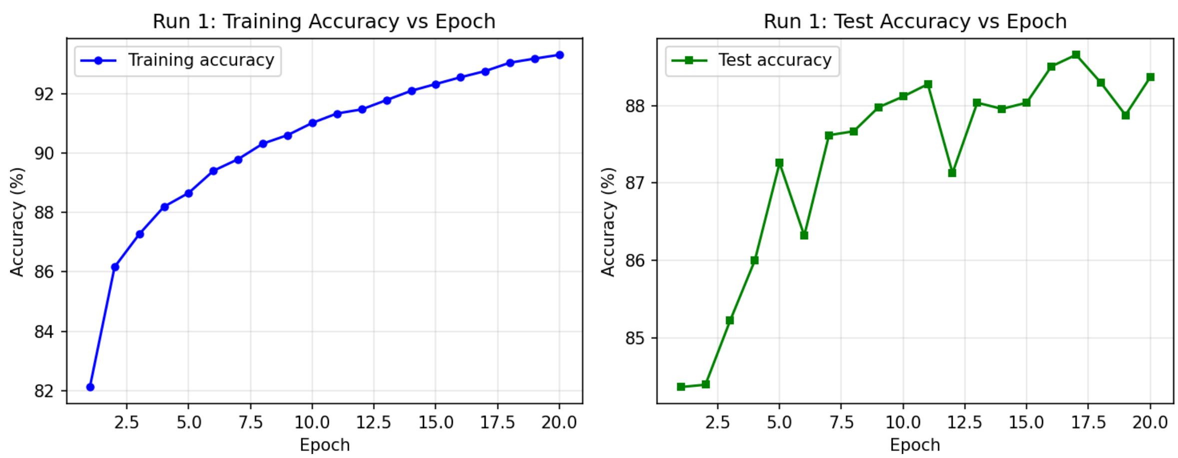

Figure 1: Centralized FashionMNIST — Run 1. Training accuracy (blue) and test accuracy (orange) vs. epoch.

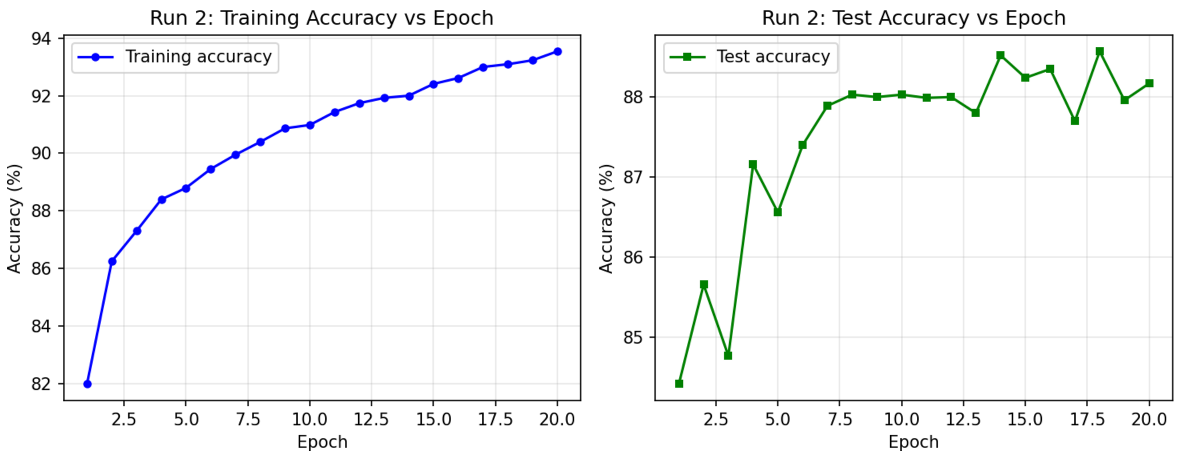

Figure 2: Centralized FashionMNIST — Run 2. Training accuracy (blue) and test accuracy (orange) vs. epoch.

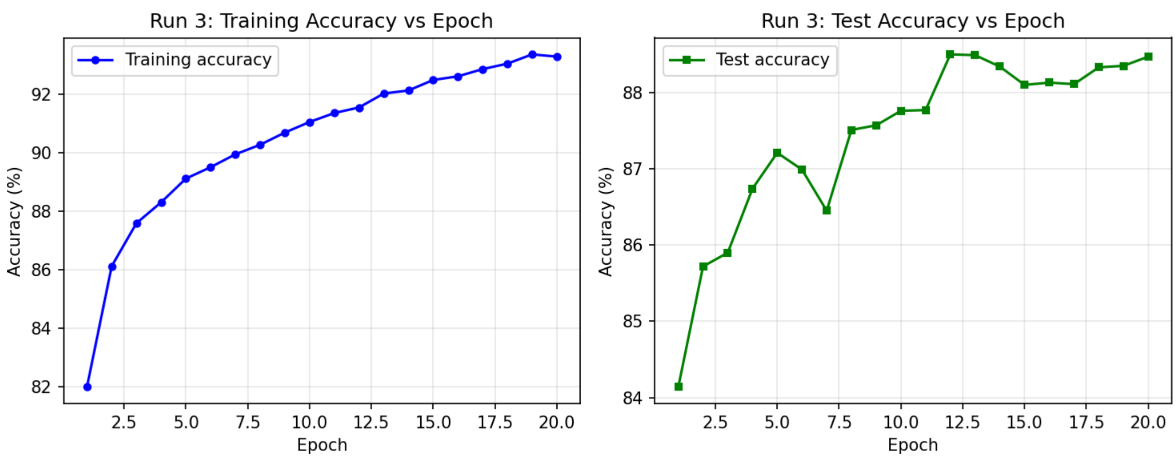

Figure 3: Centralized FashionMNIST — Run 3. Training accuracy (blue) and test accuracy (orange) vs. epoch.

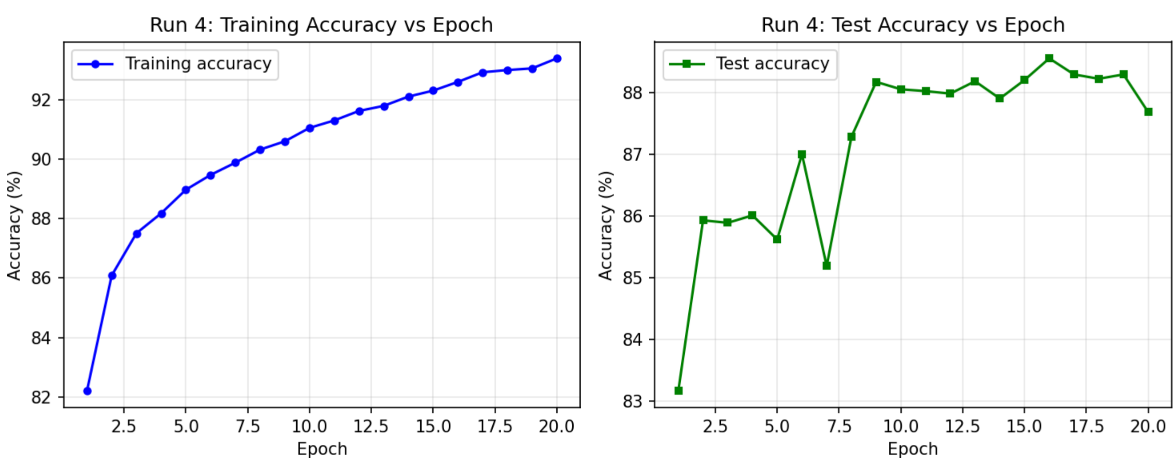

Figure 4: Centralized FashionMNIST — Run 4. Training accuracy (blue) and test accuracy (orange) vs. epoch.

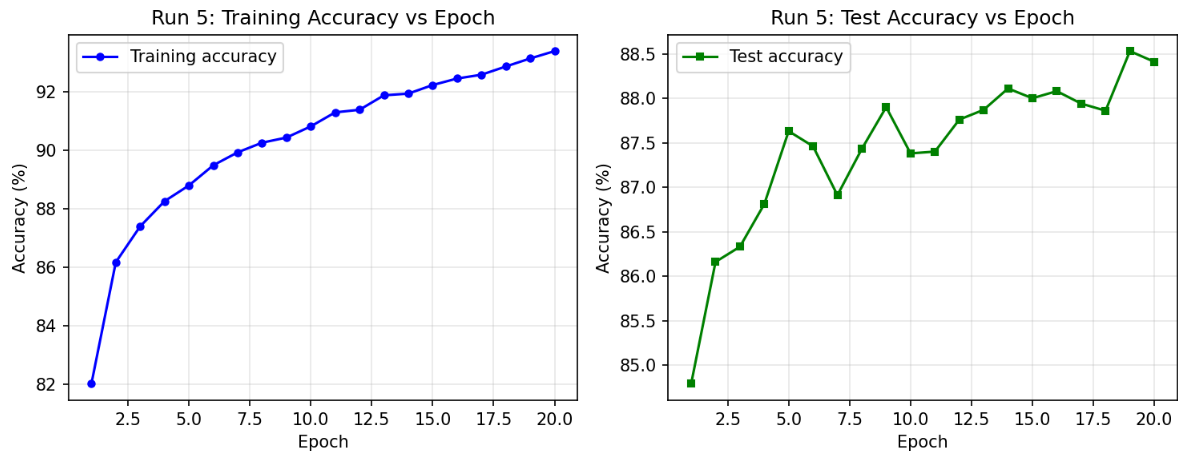

Figure 5: Centralized FashionMNIST — Run 5. Training accuracy (blue) and test accuracy (orange) vs. epoch.

3.4 Analysis

Convergence pattern: Both training and test accuracy rise steeply in epochs 1–5, then plateau. The Adam optimizer converges quickly on this relatively simple task.

- Training accuracy slightly exceeds test accuracy in all runs, indicating mild overfitting — expected for a simple MLP on a fixed dataset.
- The train-test gap remains narrow (~1–2%), confirming that the model generalizes well despite its simplicity.
- All 5 runs reach 87–89% test accuracy, demonstrating the stability of centralized training on IID data.
- **Variance = 0.0830** is the highest across all experiments. This reflects sensitivity to random initialization and batch ordering in a single-model training run — there is no averaging mechanism to smooth out lucky vs. unlucky starts.

4. Question 2 — FedAvg on FashionMNIST

4.1 Setup and Algorithm

The same MLP is now trained with FedAvg across 5 clients under a non-IID, non-overlapping data split (primary_fraction = 0.85). In each of the 20 communication rounds, the server broadcasts the global model, all clients perform 1 local epoch of SGD, and the server aggregates via weighted averaging. The global model is evaluated on the centralized test set after each round. The experiment repeats 5 times.

4.2 Results

Metric	Value
Average Final Test Accuracy (5 runs)	86.23%
Variance of Final Test Accuracy	0.0051
Gap vs. Centralized Baseline	~1.99%

4.3 Training and Test Accuracy Curves (All 5 Runs)

Each figure shows two panels: **(top)** server-side global model test accuracy per round; **(bottom)** each client's local training accuracy per round.

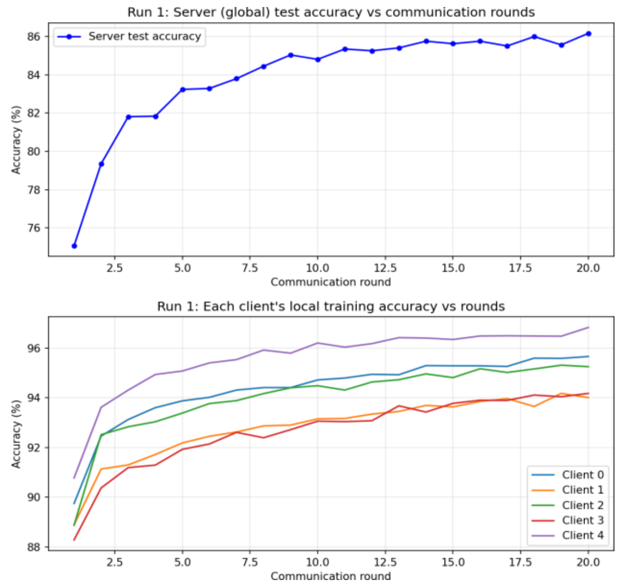

Figure 6: FedAvg FashionMNIST — Run 1. Top: global test accuracy per round. Bottom: per-client training accuracy

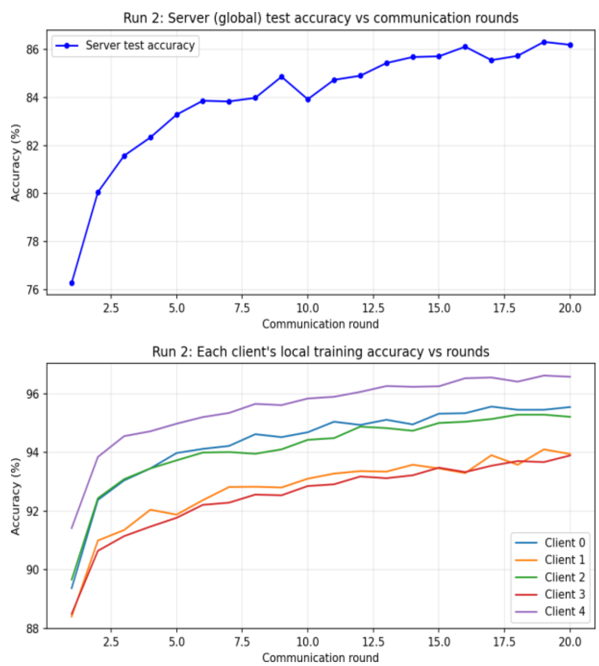

Figure 7: FedAvg FashionMNIST — Run 2. Top: global test accuracy per round. Bottom: per-client training accuracy.

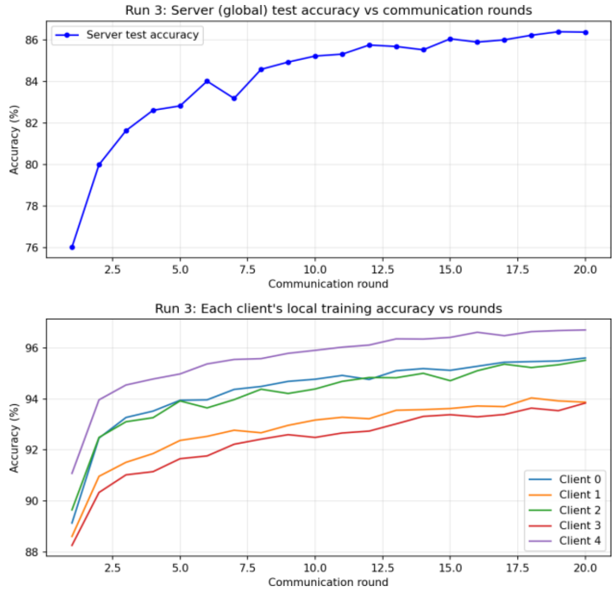

Figure 8: FedAvg FashionMNIST — Run 3. Top: global test accuracy per round. Bottom: per-client training accuracy

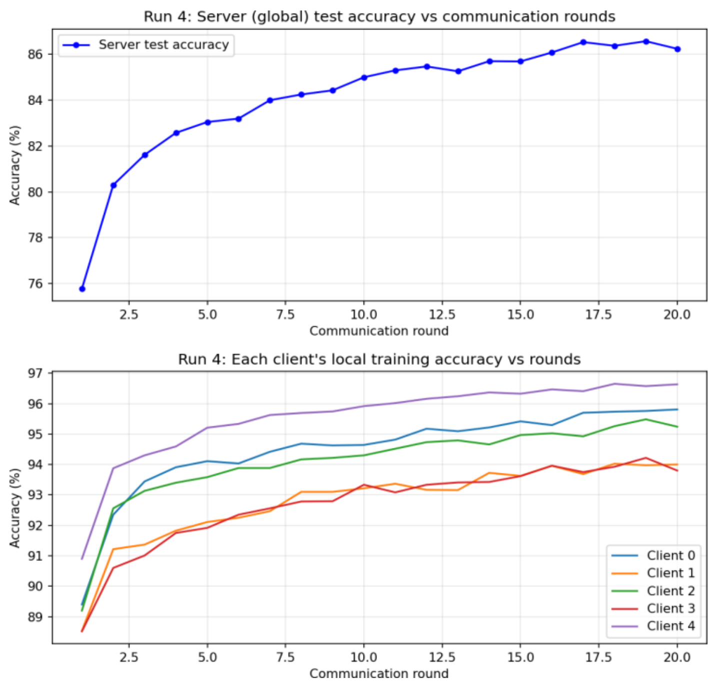

Figure 9: FedAvg FashionMNIST — Run 4. Top: global test accuracy per round. Bottom: per-client training accuracy.

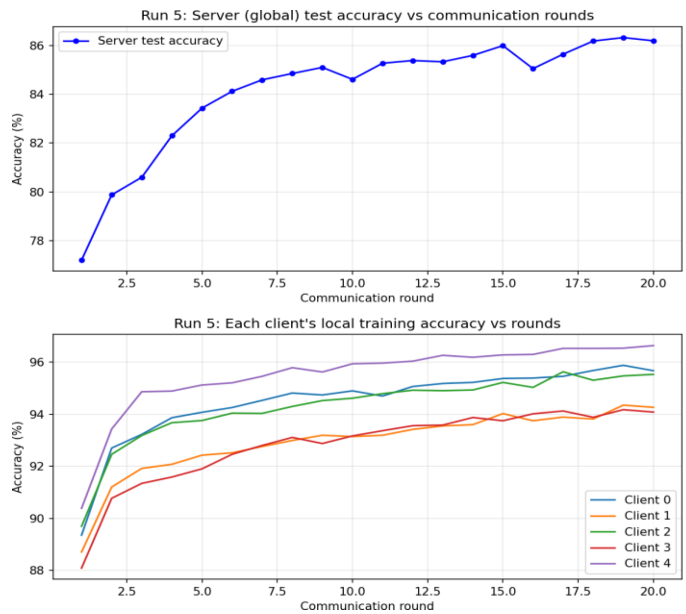

Figure 10: FedAvg FashionMNIST — Run 5. Top: global test accuracy per round. Bottom: per-client training accuracy.

4.4 Analysis

- **FedAvg recovers 86.23% accuracy** — only 1.99% below the centralized baseline — demonstrating strong performance even under non-IID conditions.
- The global test accuracy rises smoothly with communication rounds, mirroring the epoch-wise curve of centralized training.
- **Client divergence:** Per-client training curves clearly differ. Primary clients (holding ~85% of their two dominant classes) converge faster and higher locally. Non-primary clients train more slowly due to data scarcity in their non-dominant classes.
- **Variance drops from 0.0830 to 0.0051 — a 16x reduction.** Model averaging across 5 clients each round acts as an implicit regularizer, suppressing the run-to-run variability introduced by random weight initialization.
- The ~2% accuracy cost is the price of data heterogeneity and limited local computation (1 epoch/round). More rounds or local epochs could narrow this gap further.

5. Question 3 — Centralized Baseline: CIFAR-10

5.1 Why CIFAR-10 is Harder

- RGB images (3 channels vs. 1) with complex spatial structure: objects like automobiles, birds, and dogs have far greater intra-class visual variation than FashionMNIST clothing items.
- A simple CNN operates near its capacity ceiling on CIFAR-10, making training more sensitive to initialization and data distribution.
- Higher task complexity means that non-IID data splits cause greater client drift, making federated aggregation harder.

5.2 Setup

The small CNN is trained on the full CIFAR-10 training set (50,000 samples) for 20 epochs, repeated 5 independent times. Training and test accuracy are recorded per epoch.

5.3 Results

Metric	Value
Average Final Test Accuracy (5 runs)	69.84%
Variance of Final Test Accuracy	0.1803

5.4 Training and Test Accuracy Curves (All 5 Runs)

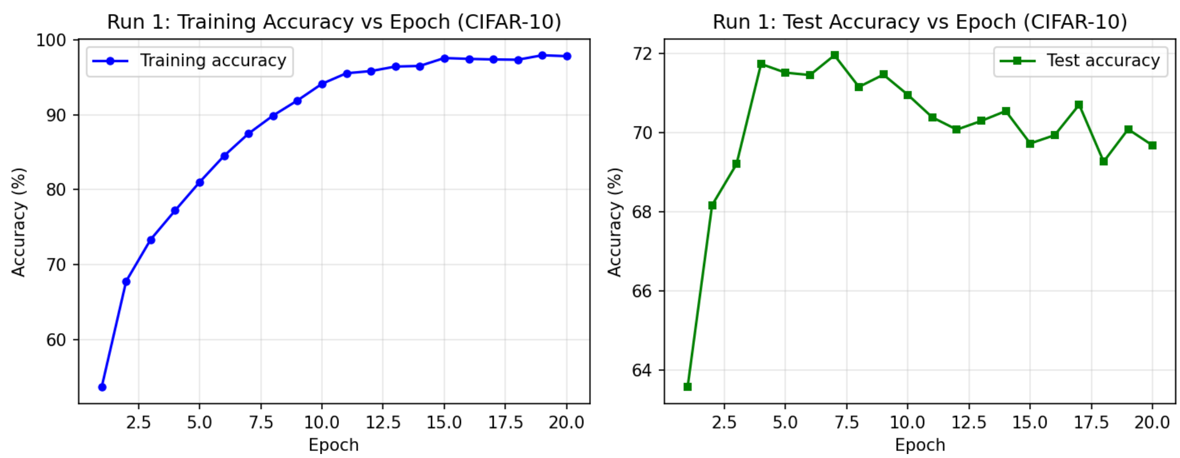

Figure 11: Centralized CIFAR-10 — Run 1. Training accuracy (blue) and test accuracy (orange) vs. epoch.

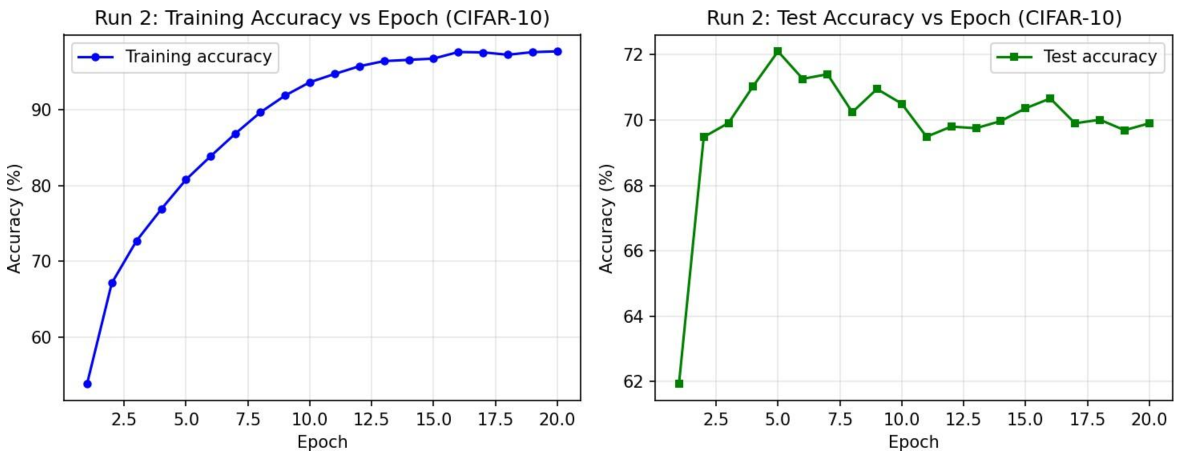

Figure 12: Centralized CIFAR-10 — Run 2. Training accuracy (blue) and test accuracy (orange) vs. epoch.

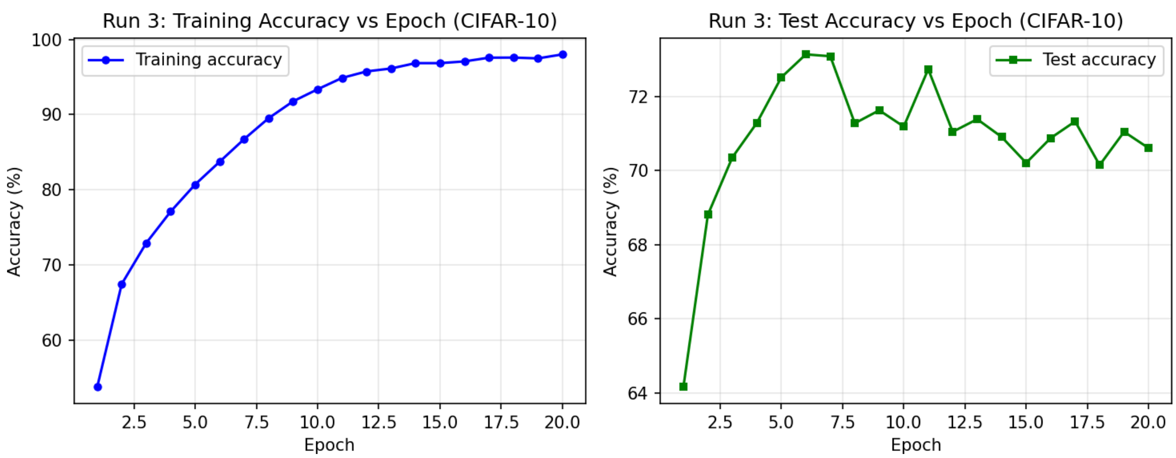

Figure 13: Centralized CIFAR-10 — Run 3. Training accuracy (blue) and test accuracy (orange) vs. epoch.

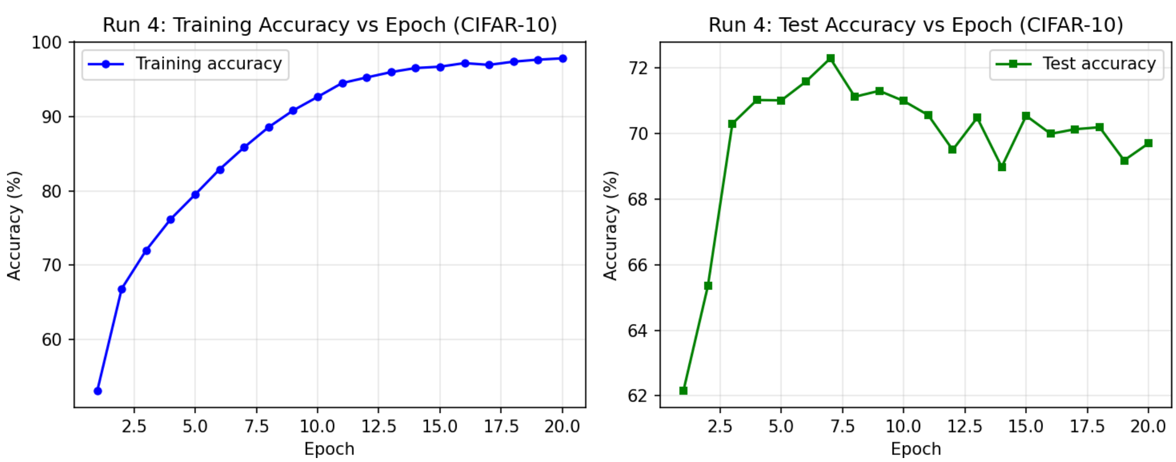

Figure 14: Centralized CIFAR-10 — Run 4. Training accuracy (blue) and test accuracy (orange) vs. epoch.

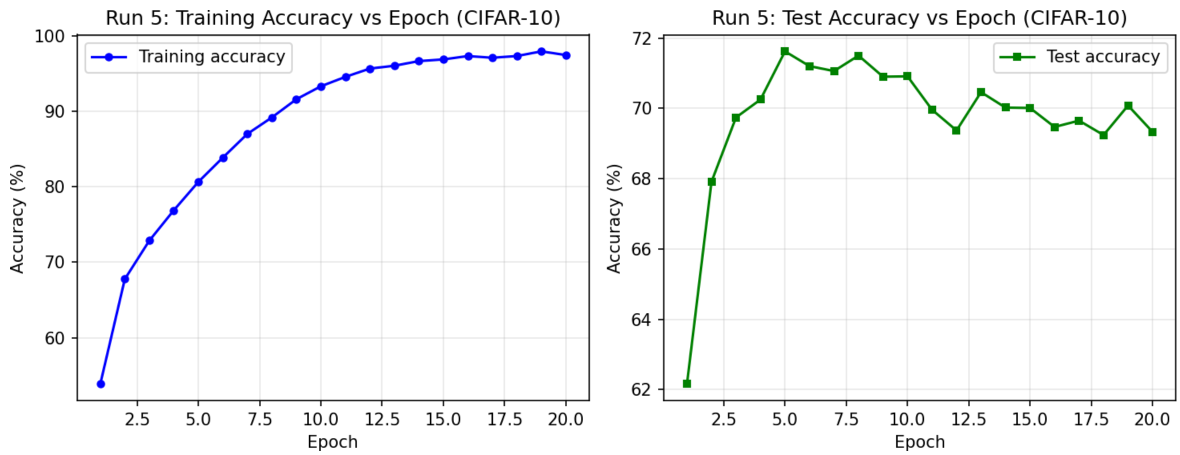

Figure 15: Centralized CIFAR-10 — Run 5. Training accuracy (blue) and test accuracy (orange) vs. epoch.

5.5 Analysis

- **69.84% accuracy** reflects the capacity limitation of the small CNN on complex RGB data. A deeper model (e.g., ResNet) would achieve higher accuracy, but the simple CNN is chosen for training speed.
- **Variance = 0.1803 is the highest across all experiments.** CIFAR-10's complex optimization landscape means different random seeds can lead to meaningfully different final solutions, especially when the model is near its capacity ceiling.
- The training-test accuracy gap is wider than FashionMNIST, indicating greater overfitting on the harder dataset.
- Training accuracy continues rising throughout 20 epochs without fully saturating, suggesting that more epochs or a larger model would help.

6. Question 3 — FedAvg on CIFAR-10

6.1 Setup

The same CNN is trained with FedAvg under a **stronger non-IID setting**: primary_fraction = 0.95. Each primary client retains 95% of its two assigned classes, leaving only 5% for other clients. This creates highly skewed local distributions where clients are near-specialists on 2 of the 10 classes.

The 0.95 setting is deliberately chosen (over 0.85) to produce a larger, more illustrative accuracy gap on the harder CIFAR-10 dataset, enabling a cleaner comparison with FashionMNIST results.

6.2 Results

Metric	Value
Average Final Test Accuracy (5 runs)	66.07%
Variance of Final Test Accuracy	0.0521
Gap vs. Centralized Baseline	~3.77%

6.3 Training and Test Accuracy Curves (All 5 Runs)

Each figure shows two panels: **(top)** global test accuracy per round; **(bottom)** per-client training accuracy per round.

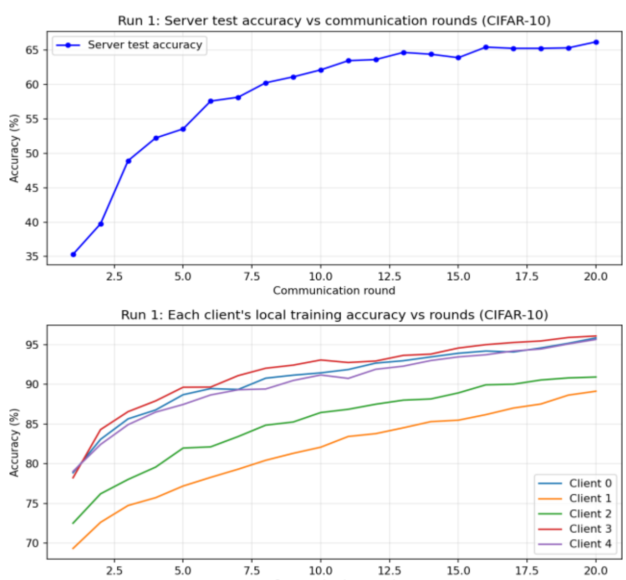

Figure 16: FedAvg CIFAR-10 (0.95) — Run 1. Top: global test accuracy per round. Bottom: per-client training accuracy.

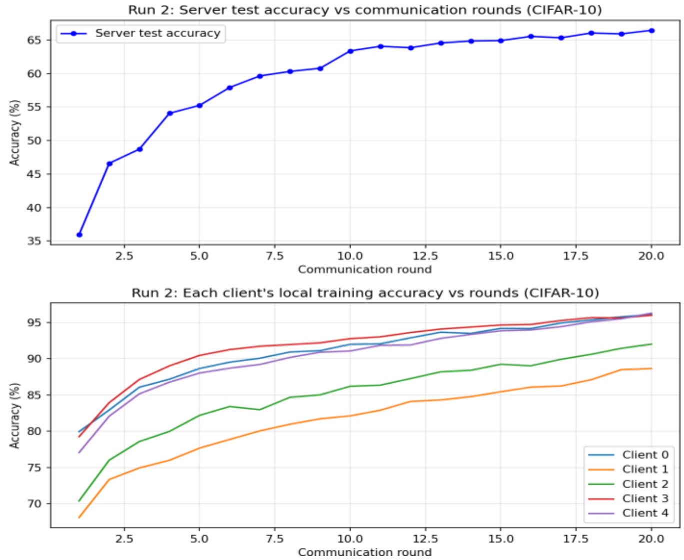

Figure 17: FedAvg CIFAR-10 (0.95) — Run 2. Top: global test accuracy per round. Bottom: per-client training accuracy.

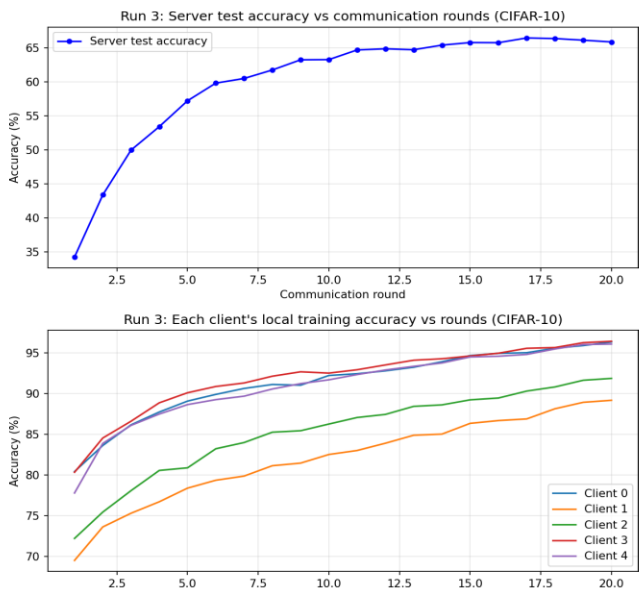

Figure 18: FedAvg CIFAR-10 (0.95) — Run 3. Top: global test accuracy per round. Bottom: per-client training accuracy.

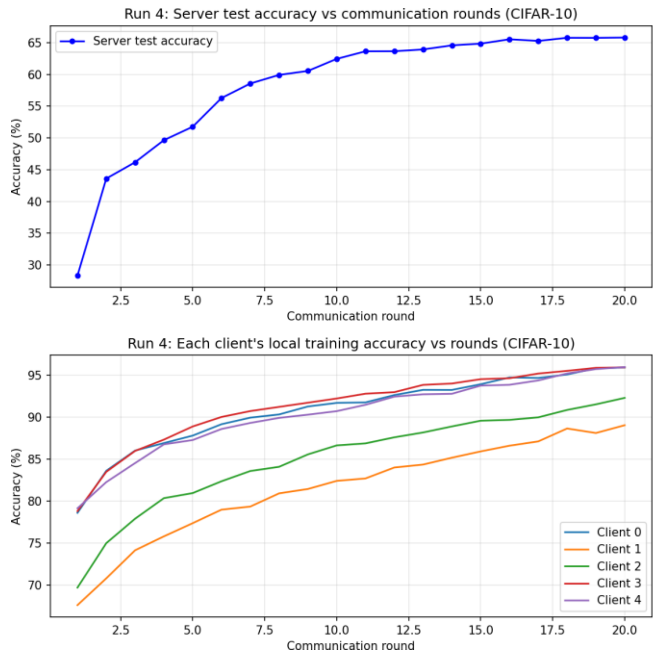

Figure 19: FedAvg CIFAR-10 (0.95) — Run 4. Top: global test accuracy per round. Bottom: per-client training accuracy.

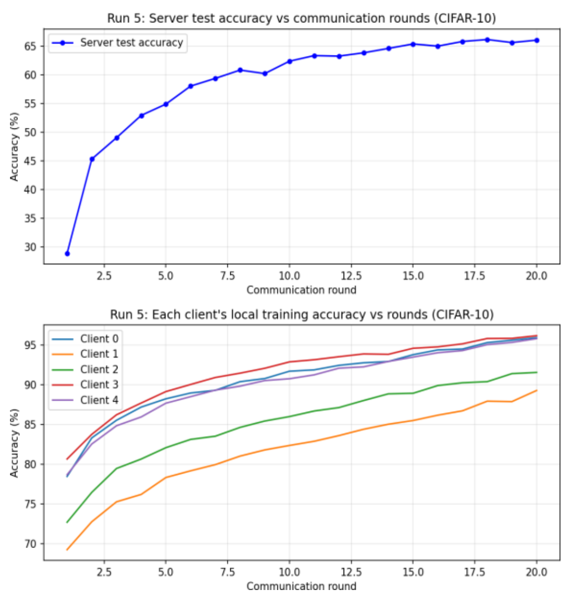

Figure 20: FedAvg CIFAR-10 (0.95) — Run 5. Top: global test accuracy per round. Bottom: per-client training accuracy.

6.4 Analysis

- **3.77% accuracy gap** is larger than FashionMNIST's 1.99%, confirming that stronger non-IID settings on harder tasks impose a greater federated learning penalty.
- Per-client training curves show more extreme divergence: clients strongly specialize to their 2 dominant classes. Client 0 (classes 0,1) and Client 4 (classes 8,9) may achieve high local accuracy but their models are heavily biased.
- **Despite the strong non-IID setting, FedAvg still achieves 66.07%** — demonstrating the algorithm's robustness. Weighted averaging successfully reconciles locally biased models into a reasonably competent global model.
- Variance drops from 0.1803 (centralized) to 0.0521, again showing the regularizing effect of model

averaging across clients.

7. Comparative Analysis and Conclusions

7.1 Summary Table

Setting	Avg Test Acc.	Variance	Gap vs Central	Var Reduction
Centralized FashionMNIST	88.22%	0.0830	—	—
FedAvg FashionMNIST	86.23%	0.0051	1.99%	16x lower
Centralized CIFAR-10	69.84%	0.1803	—	—
FedAvg CIFAR-10 (0.95)	66.07%	0.0521	3.77%	3.5x lower

7.2 Observation 1: FedAvg achieves slightly lower accuracy than centralized training

Across both datasets, FedAvg achieves lower final test accuracy than its centralized counterpart. The root cause is **client drift**: because each client's data distribution is biased toward its dominant classes, local SGD updates move model weights in directions that are suboptimal for the global objective. When the server averages these drifted models, the result is a compromise that is worse than training on the full IID dataset.

The gap is small for FashionMNIST (1.99%) because the task is simple enough that even biased local models learn broadly generalizable features. For CIFAR-10 (3.77%), the greater task complexity amplifies client drift — local models that have seen only ~5% of non-dominant classes cannot accurately classify them.

7.3 Observation 2: FedAvg Dramatically Reduces Variance

FedAvg consistently produces **lower run-to-run variance** than centralized training:

- **FashionMNIST:** variance drops from 0.0830 to 0.0051 — a 16x reduction.
- **CIFAR-10:** variance drops from 0.1803 to 0.0521 — a 3.5x reduction.

This is the **implicit ensemble effect** of FedAvg: aggregating 5 independently initialized and trained local models each round smooths out noise introduced by any single random seed. Centralized training has no such mechanism — one unlucky initialization can produce a noticeably weaker final model.

7.4 Observation 3: Stronger Non-IID Increases the Accuracy Gap

Comparing FashionMNIST (0.85) and CIFAR-10 (0.95), the stronger non-IID setting on the harder dataset produces a larger gap. This is theoretically expected: as the primary fraction approaches 1.0, each client's local model converges to a near-optimal classifier for its 2 classes but a near-random classifier for the other 8. Averaging these highly specialized models produces a mediocre global model.

7.5 Observation 4: Dataset Complexity Amplifies All Effects

- Absolute accuracy is lower for CIFAR-10 (69.84% vs 88.22% centralized) due to greater visual complexity and limited model capacity.
- Variance is higher for CIFAR-10 in both settings — a more rugged loss landscape makes training outcomes more sensitive to initialization.
- The federated penalty is larger for CIFAR-10 — harder tasks are more susceptible to the data heterogeneity induced by non-IID partitioning.

7.6 Key Conclusions

- **FedAvg is effective:** It recovers the majority of centralized performance (86.23% vs 88.22% on FashionMNIST) despite non-IID data, communication constraints, and no data sharing. For privacy-sensitive applications, this trade-off is acceptable.
- **The accuracy-variance trade-off:** FedAvg trades a small reduction in mean accuracy for a significant reduction in variance. This makes federated models more predictably reliable across deployment scenarios.
- **Non-IID data is the primary challenge:** The degree of data skew directly controls the performance gap. Techniques like FedProx, SCAFFOLD, or data-free knowledge distillation are designed specifically to mitigate this.
- **Task complexity matters:** FedAvg performs better (relative to centralized) on simpler tasks. For complex tasks (CIFAR-10 and beyond), more rounds, more local epochs, or better aggregation strategies are needed.
- **Communication efficiency:** 20 rounds suffice to approach convergence on FashionMNIST. This is consistent with McMahan et al.'s finding that FedAvg reduces required communication rounds by 10-100x compared to naive distributed SGD.

8. References

[1] McMahan, H. B., Moore, E., Ramage, D., Hampson, S., & Agüera y Arcas, B. (2017). Communication-efficient learning of deep networks from decentralized data. *AISTATS 2017*, JMLR: W&CP volume 54. arXiv:1602.05629.

[2] Xiao, H., Rasul, K., & Vollgraf, R. (2017). Fashion-MNIST: A novel image dataset for benchmarking machine learning algorithms. *arXiv preprint arXiv:1708.07747*.

[3] Krizhevsky, A. (2009). Learning multiple layers of features from tiny images. *Technical Report, University of Toronto.*

[4] Li, T., Sahu, A. K., Zaheer, M., Sanjabi, M., Talwalkar, A., & Smith, V. (2020). Federated optimization in heterogeneous networks (FedProx). *Proceedings of Machine Learning and Systems, 2*, 429-450.

[5] Kairouz, P., McMahan, H. B., et al. (2021). Advances and open problems in federated learning. *Foundations and Trends in Machine Learning, 14*(1-2), 1-210.